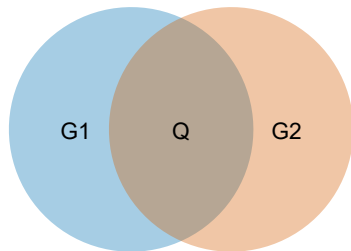


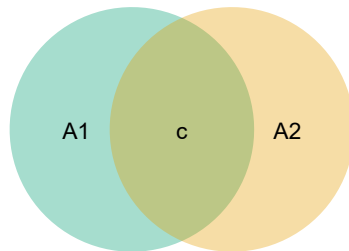
A



$$P1=0.87 \quad P2=0.85 \quad Q=0.80$$

$$J_{\text{old}} = Q/(P1+P2-Q) = 0.86$$

B



$$a=0.86 \quad b=0.83 \quad c=0.74$$

new gate: G and A

C

$$J_{\text{new}} = Qc/(P1a+P2b-Qc)$$

$$= 0.68 \quad (<= J_{\text{old}} = 0.86)$$

$$J_{\text{new}} <= J_{\text{old}} \quad \text{iff} \quad c(P1+P2) <= P1a+P2b$$

$$(1.27 <= 1.45)$$

symmetric: $a=b=p, c \leq p \Rightarrow$ non-increasing